COLIN NGUYEN

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ML researcher and AI engineer working on evaluation under uncertainty: how LLM agents and health-sensing models are measured, and how to train models efficiently. Background in statistics; co-inventor on a pending AI patent.

RESEARCH EXPERIENCE

Evaluation of Mobile-Health Receptivity Models — with Dr. Roberto Benzo (OSU), Dr. Peter Washington (UCSF) *2026 – Present*

- Showed that pooled AUC, the standard metric for models predicting when someone will answer a health prompt, is dominated by between-person comparisons: 99.11–99.73% of scored pairs compare different people across three cohorts (n = 63, 329, 351).
- Pre-registered a personalisation study (OSF) and trained a self-supervised GRU encoder whose embeddings identify participants far above baseline (0.4482 vs 0.0165) but sit at chance for within-person response.

Efficient Training Methods for Language Models — with Prof. Eric Suess, CSU East Bay *2026 – Present*

- Measured five memory-saving techniques for fine-tuning SmolLM2-135M/360M in PyTorch, raising the trainable step from 8,192 to 131,072 tokens in 71% of baseline memory.
- Wrote a block-wise 8-bit AdamW from scratch that matched torchao's optimizer state to the byte and was statistically equivalent to fp32 AdamW in final eval loss across six seeds (pre-registered TOST equivalence test).

The Ohio State University — The Lifestyles Lab (PI: Dr. Roberto Benzo) *2026 – Present*

- Co-leading the analysis plan for validating Fitbit sleep measurement against research-grade actigraphy (ActiGraph) in cancer survivors: night-matched Bland–Altman analysis, mixed models, and TOST equivalence testing.
- Migrated the lab's data-collection tool from the retiring Fitbit API to the Google Health API, with a fault-injected parity harness that caught silent data defects.

Systematic Review: Computer Vision for Home-Video Phenotyping of ASD and ADHD — UCSF *Oct 2025 – Present*

- Screened studies for a ~55-study review (lead: Dr. Abdur Rasool; PI: Dr. Peter Washington); responsible for the meta-regression of model performance on video length, setting, external validation, and ground-truth source. *Manuscript in preparation.*

INDUSTRY EXPERIENCE

Athena Money — AI Engineer *2025 – Present*

- Designed a multi-turn LLM evaluation platform for agents in regulated mortgage underwriting, with three evaluator types measuring different things: LLM-as-judge rubrics, per-turn quality scoring, and outcome checkpoints, driven by an LLM user-simulator.
- Handled run-to-run variance with field-level consensus across N trials, confidence-score calibration, and statistical regression tests; built the underwriting agents under evaluation (Google ADK, Temporal).

Zeta Global — Data Science & GenAI Intern *2025*

- Multi-agent LLM orchestration (LangGraph) and an evaluation framework (accuracy / consistency / latency / cost); co-inventor on pending patent "Adaptive Neural Content Generation."

NationBuilder — AI Research Intern *2025*

- RAG pipeline (MiniLM embeddings, FAISS) with 93% relevance-classification accuracy, and agentic document extraction for civic data.

SELECTED PROJECTS

Inverse Optimization of NBA Shot Decisions — UCSB *2025*

- Formulated shot selection as inverse optimization over 128,069 shots; a PyTorch cost network with KKT-condition penalties recovers each player's latent cost function, with defender distance the top factor for all three players studied.

EDUCATION

University of California, Santa Barbara *Sep 2022 – Jun 2025*

B.S. in Statistics & Data Science, Minor in Mathematics *Santa Barbara, CA*

Relevant coursework: Operations Research, Advanced Linear Algebra I–II, Statistical Machine Learning, Advanced Statistical Models, Regression Analysis, Big Data Analytics, Partial Differential Equations.

SKILLS

Research methods: evaluation methodology, equivalence testing (TOST, Bland–Altman), calibration, pre-registration, experimental design, inverse optimization

ML / DL: PyTorch, fine-tuning / PEFT (LoRA, QLoRA), vLLM, scikit-learn, TensorFlow, Node2Vec, HuggingFace Transformers

LLM systems: RAG, LLM evaluation, LangGraph, embeddings, vector search (FAISS)

Languages / Tools: Python, R, SQL, Java, Go; AWS, Google Cloud, Docker, Spark, Databricks

PATENTS

- "Adaptive Neural Content Generation" — U.S. Provisional Patent Application No. 63/943,780, filed Dec 2025; Zeta Global (first-named inventor, with D. Portman).